

# CompositionProof-2 Governance Protocol

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## Authority Model: FRESH GO EACH CYCLE

Established 2026-09-08 by Derek Hone, CEO. Applies to all future CompositionProof-2 epoch submissions.

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## Epoch Submission Protocol

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**Do NOT create standing authorization.** Each consequential hardware execution under a new calibration state requires authority re-established at the moment of execution, not inherited from prior GO.

For every future CompositionProof-2 epoch:

1. **Verify** genuinely new `ibm_marrakesh` calibration timestamp
  - ( `backend.properties().last_update_date` )
  - Compare against all prior epochs in `epoch_series.json`
  - If timestamp matches any existing epoch, STOP - not a distinct calibration state
2. **Verify** frozen manifest remains intact
  - Run: `python3 build_lock.py --verify`
  - Composite hash must match: `ebe0919b997341569670ccd27b2a8ff754599d242489a2b06ff4d-b20c8872fd6`
  - If verification fails, STOP - do not submit on altered design
3. **Report** to Derek before any hardware submission:
  - Current QPU allocation (consumed / remaining / total, 28-day window dates)
  - Prior epoch count (N of  $\geq 20$  target)
  - New calibration timestamp vs most recent prior epoch timestamp
  - Manifest verification status (INTACT / ALTERED)
4. **STOP** and await explicit authorization
  - Required token: **"GO COMPOSITIONPROOF-2 EPOCH N"** where N = next epoch number
  - No hardware submission until Derek explicitly replies with this token
5. **Submit** only after receiving explicit GO
  - Exactly one epoch battery (8 circuits × 4000 shots)
  - Use `submit_epoch.py` with `--go-token "GO COMPOSITIONPROOF-2" --i-have-ceo-authorization --epoch-id N --backend ibm_marrakesh`
6. **Preserve** every epoch regardless of result
  - Record to `epoch_series.json`
  - Create `EPOCHN_RECORD.md` (+ .pdf/.docx)
  - Git commit
  - No deletion, no polishing, no reruns of same calibration state
7. **Verdict discipline**
  - No final CP-2 verdict before  $N_{MIN}=20$  distinct calibration epochs
  - Epoch-level records state "COLLECTED - NO FINAL VERDICT"

- Only after  $\geq 20$  epochs: compute Spearman  $\rho_s$ , bootstrap CI, apply classification rules per `lock_candidate.json`

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## Rationale (Derek Hone, 2026-09-08)

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Each CP-2 epoch is a new consequential hardware execution under a new calibration state, so authority should be re-established at the moment of execution rather than inherited from Epoch 1. It also gives you a chance to review remaining QPU budget, calibration timestamp, prior-epoch integrity, and whether anything in the campaign has changed before another irreversible submission.

**Proof Before Power applied to the experiment itself:** prior authorization does not automatically become current execution permission.

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## Current State

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- **Epoch 1:** COLLECTED 2026-09-08, job `dag4fnfi3e6s738ltnpg`, cal `2026-09-08T16:59:31+00:00`, S=2.7205, M=3.7710, 11s QPU
  - **Next epoch:** Awaits fresh `ibm_marrakesh` recalibration + explicit GO COMPOSITIONPROOF-2 EPOCH 2
  - **QPU remaining:** ~352s of 600s (as of 2026-09-08, window 2026-08-11 to 2026-09-09)
  - **Manifest lock:** INTACT `ebe0919b...72fd6` (verified pre-Epoch-1)
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**Protocol status: ACTIVE**